

Country Development Clustering: Identifying Nations in Need of Humanitarian Aid

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Abstract

This report presents a clustering-based analysis of 167 countries using socio-economic and health indicators to identify nations most in need of financial aid. Principal Component Analysis (PCA) was applied to reduce dimensionality, followed by K-Means, hierarchical, and DBSCAN clustering. A critical methodological decision was the removal of 26 outlier countries identified via the IQR method on PCA components prior to clustering. However, recognizing that outlier removal could inadvertently exclude vulnerable nations, a separate investigation was conducted on the removed observations. The results show that countries such as Haiti and the Central African Republic were among the removed outliers but ranked highest in aid priority. This finding underscores the importance of careful outlier handling in real-world data science projects, where overlooking such cases could lead to misleading conclusions and misallocated resources.

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1 Introduction

International development organizations face the critical challenge of allocating limited humanitarian resources to countries with the greatest need. Data-driven approaches can help identify which nations require urgent financial assistance by analyzing socio-economic and health indicators.

This project analyzes a dataset of 167 countries using nine development indicators: child mortality, exports, health spending, imports, income, inflation, life expectancy, fertility rate, and GDP per capita. The goal is to group countries by their development profile and recommend the top candidates for humanitarian aid.

The full analysis code is available on GitHub: <https://github.com/fermow/SBU-Data-Science-20>

2 Data and Methods

2.1 Dataset

The dataset contains 167 countries with 10 variables:

Variable	Type	Description
country	Categorical	Country name
child_mort	Float	Death of children under 5 per 1000 live births
exports	Float	Exports per capita (% of GDP per capita)
health	Float	Health spending per capita (% of GDP per capita)
imports	Float	Imports per capita (% of GDP per capita)
income	Integer	Net income per person
inflation	Float	Annual inflation rate (%)
life_expec	Float	Life expectancy at birth (years)
total_fer	Float	Total fertility rate (children per woman)
gdpp	Integer	GDP per capita

Table 1: Description of dataset variables

2.2 Preprocessing

Three variables (exports, imports, health) were originally expressed as percentages of GDP. These were converted to absolute values by multiplying by GDP per capita:

$$\text{variable} = \text{variable_pct} \times \frac{\text{gdpp}}{100}$$

All numeric features were standardized using StandardScaler to have zero mean and unit variance, ensuring that variables with larger scales do not dominate distance calculations.

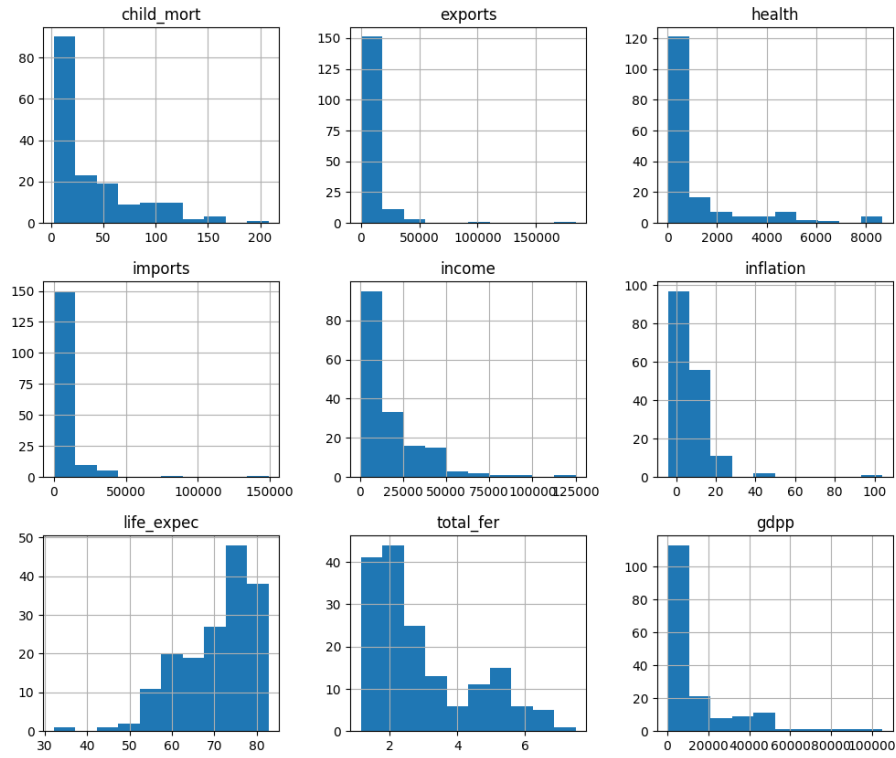


Figure 1: Distribution of original features before standardization. Significant differences in scale and skewness are visible.

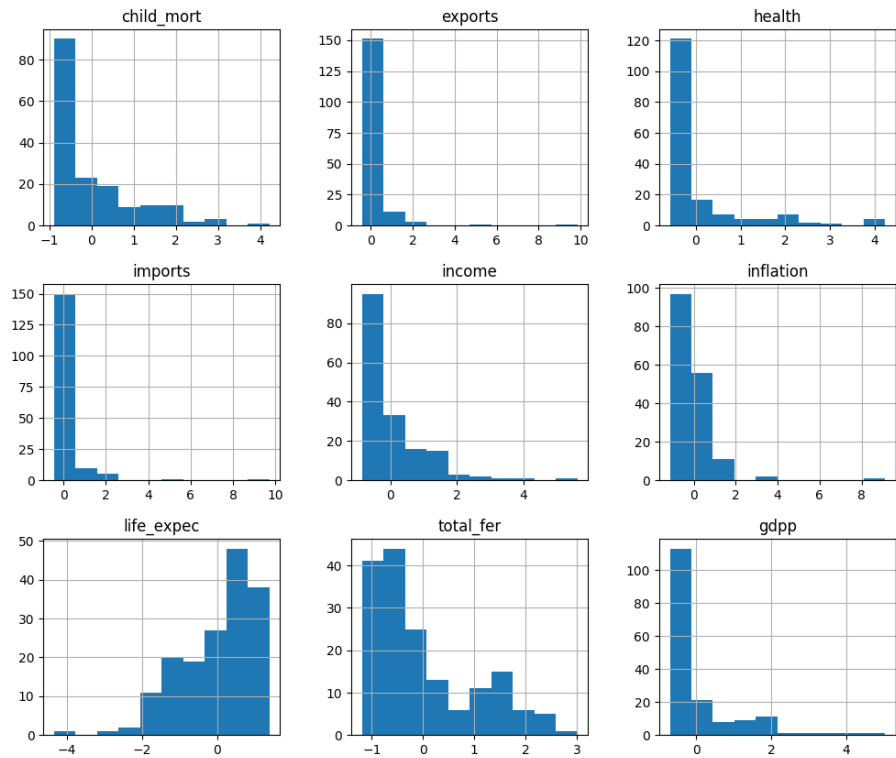


Figure 2: Distribution of standardized features. All variables now have zero mean and unit variance.

2.3 Correlation Analysis

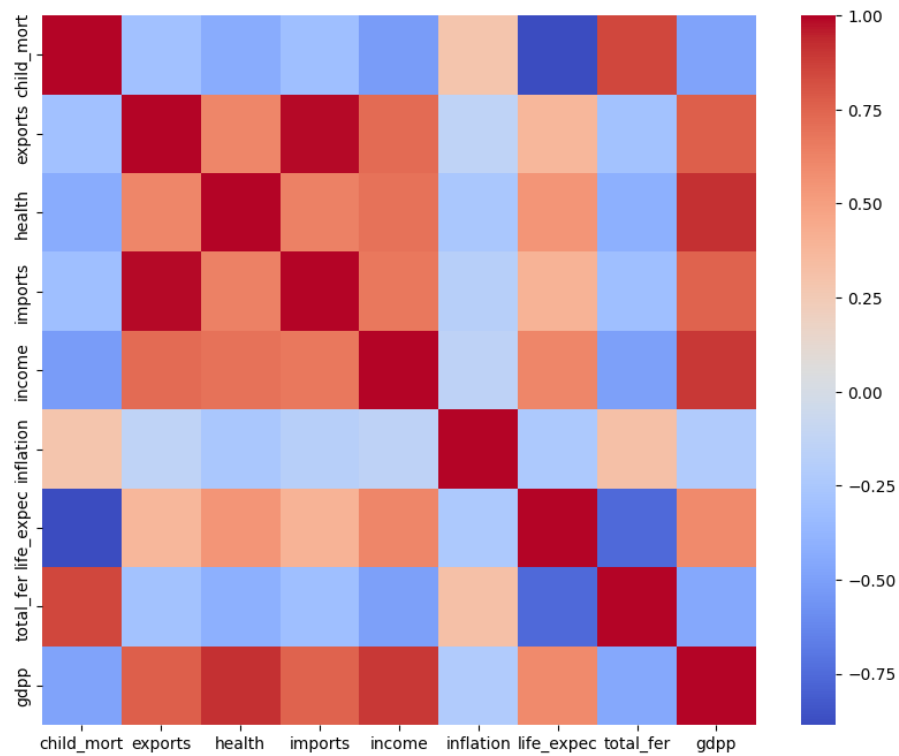


Figure 3: Correlation heatmap of features. Strong positive correlations exist between income and GDP per capita, and between child mortality and fertility rate.

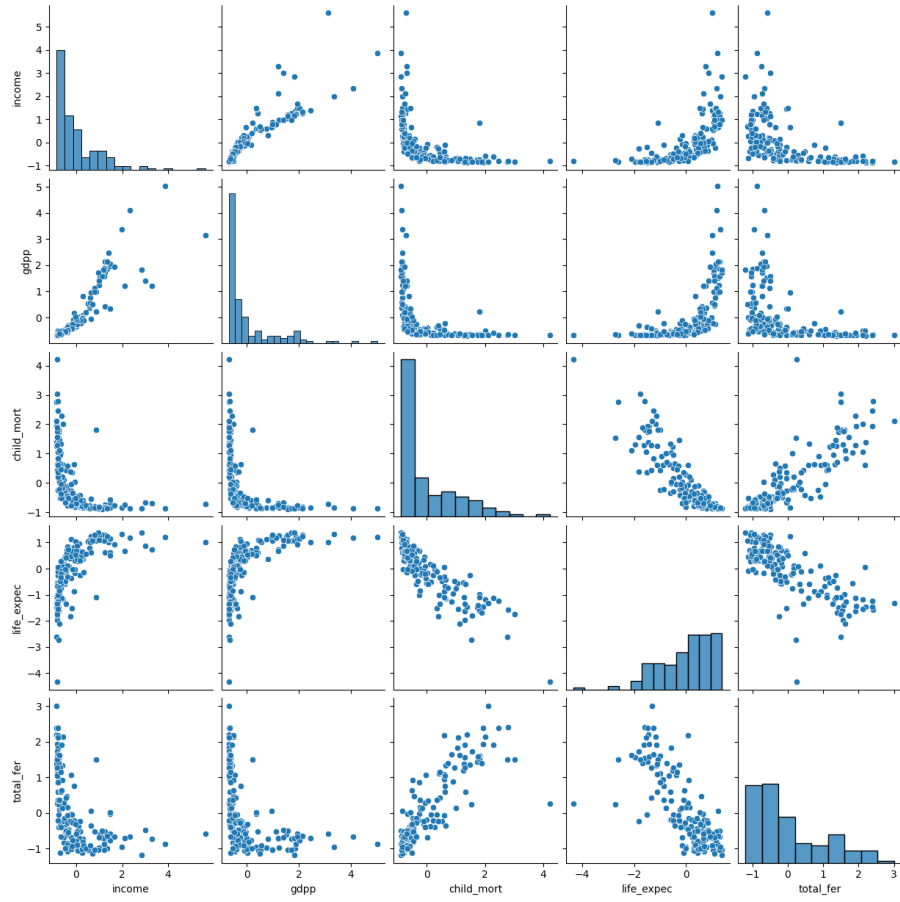


Figure 4: Pairplot of key development indicators showing relationships between income, GDP per capita, child mortality, life expectancy, and fertility rate.

2.4 Outlier Analysis on Raw Features

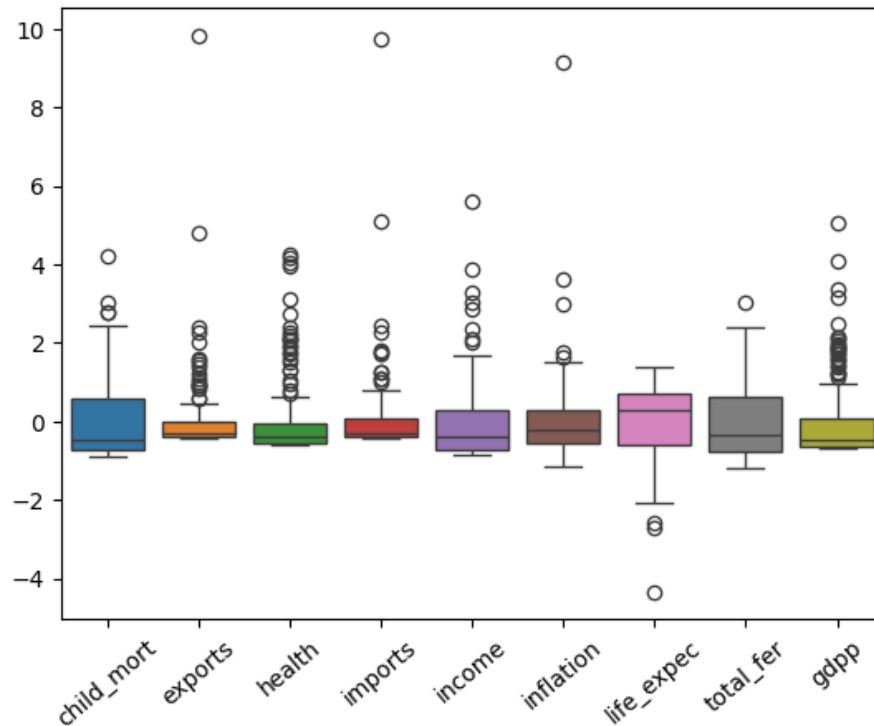


Figure 5: Boxplot of standardized features before PCA. Multiple variables exhibit extreme values, particularly exports, imports, and GDP per capita.

2.5 Dimensionality Reduction

Principal Component Analysis (PCA) was applied to the standardized data. The first four principal components explain over 93% of the total variance:

- PC1: 58.94%
- PC2: 18.45%
- PC3: 9.91%
- PC4: 6.07%
- Cumulative: 93.37%

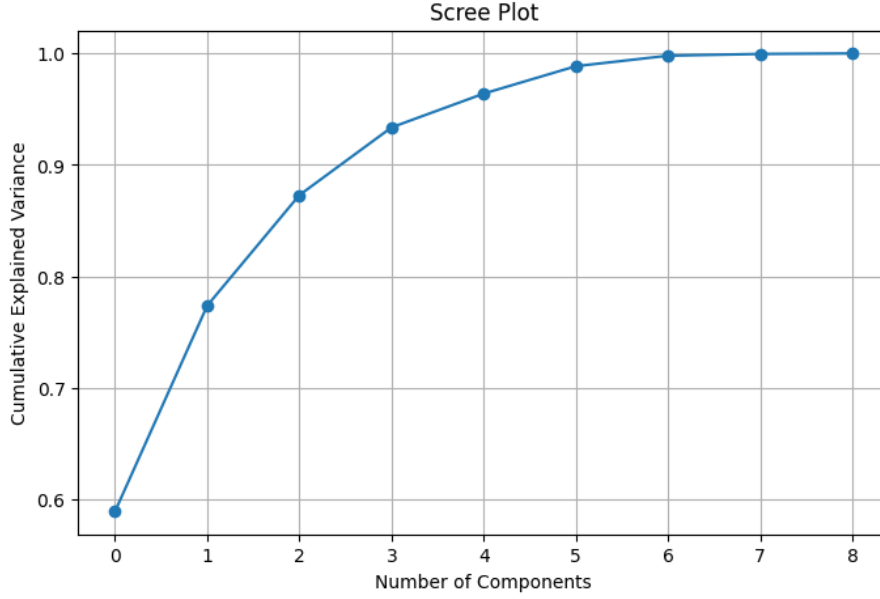


Figure 6: Scree plot showing cumulative explained variance by principal components. Four components were retained.

The dataset was transformed into four principal components (PC1–PC4) for clustering.

2.6 Outlier Detection on PCA Components

Boxplot analysis on the principal components revealed extreme observations. Using the interquartile range (IQR) method ($1.5 \times \text{IQR}$ rule), 26 countries were identified as outliers and removed from the main clustering pipeline.

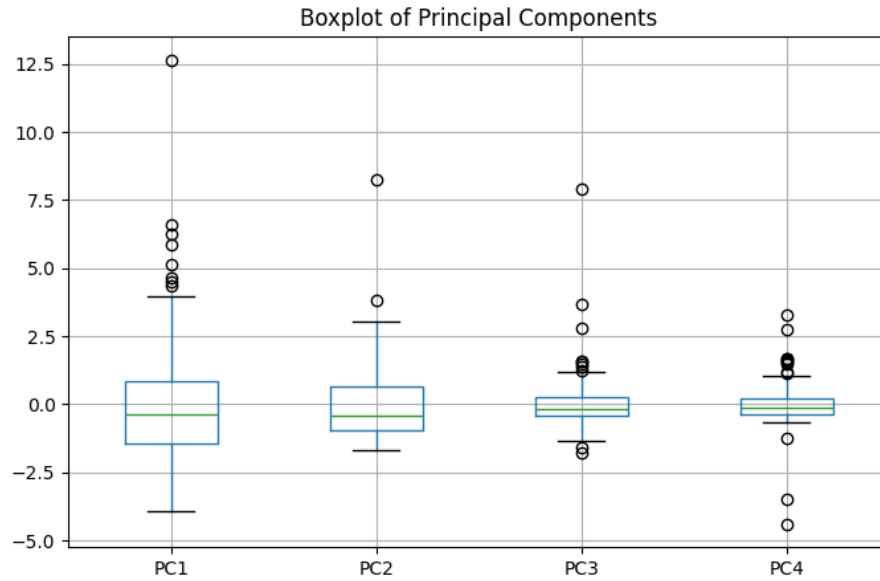


Figure 7: Boxplot of principal components before outlier removal. Several extreme values are visible.

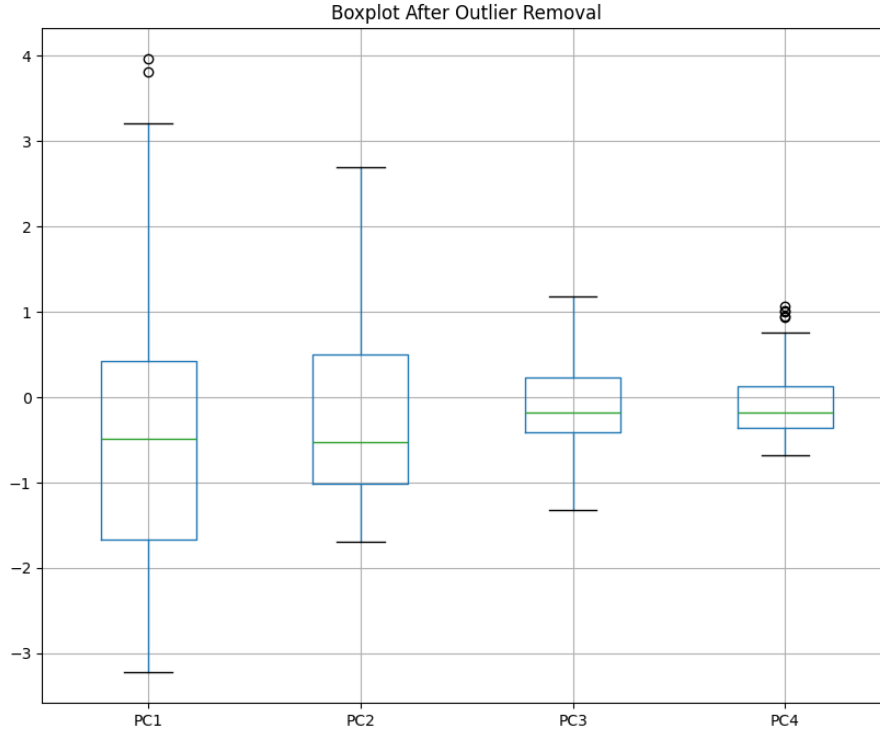


Figure 8: Boxplot of principal components after outlier removal. The distribution is more compact.

The 26 removed outlier countries were:

Argentina, Australia, Brunei, Canada, Central African Republic, Denmark, France, Germany, Haiti, Ireland, Japan, Luxembourg, Malta, Mongolia, Netherlands, Nigeria, Norway, Qatar, Saudi Arabia, Singapore, Sri Lanka, Sweden, Switzerland, Timor-Leste, United States, Venezuela

These include both highly developed economies (e.g., Switzerland, Norway, United States) and countries with severe development challenges (e.g., Haiti, Central African Republic, Nigeria, Timor-Leste).

3 Clustering Results

3.1 Optimal Number of Clusters

Both the elbow method and silhouette analysis were evaluated for k ranging from 2 to 10.

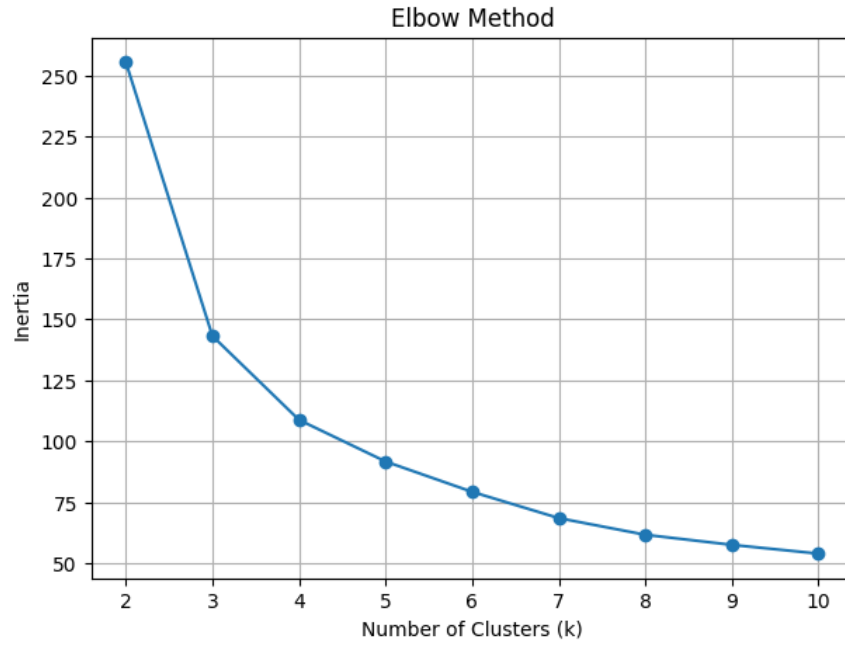


Figure 9: Elbow method for K-Means. The elbow point suggests $k = 3$ or $k = 4$.

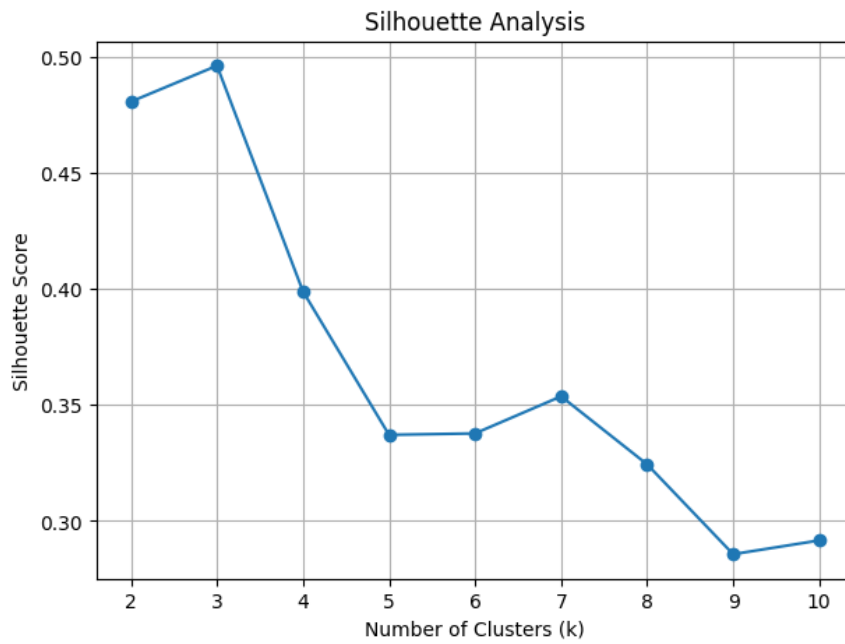


Figure 10: Silhouette scores for different values of k . The highest score occurs at $k = 3$.

Based on the silhouette score (highest at $k = 3$) and interpretability, three clusters were selected.

3.2 K-Means Clustering ($k = 3$)

Variable	Cluster 0	Cluster 1	Cluster 2
Count	21	76	44
child_mort	5.75	22.42	87.45
exports	15073.75	2616.79	940.28
health	2557.89	383.48	113.67
imports	13801.20	2858.42	857.83
income	36866.67	11405.53	4038.30
inflation	3.19	6.19	9.86
life_expec	79.14	72.87	59.83
total_fer	1.83	2.27	4.98
gdpp	31080.95	5890.95	1922.82

Table 2: Cluster characteristics (mean values) for K-Means with $k = 3$.

Cluster 0 (21 countries): Highly developed nations with the highest income, GDP per capita, life expectancy, and lowest child mortality and fertility rates.

Cluster 1 (76 countries): Moderately developed countries with mid-range indicators.

Cluster 2 (44 countries): The most disadvantaged group, characterized by the highest child mortality (mean = 87.45), lowest income (mean = \$4,038), lowest GDP per capita (mean = \$1,923), lowest life expectancy (mean = 59.83 years), and highest fertility rates (mean = 4.98). These countries were identified as the highest priority candidates for humanitarian aid.

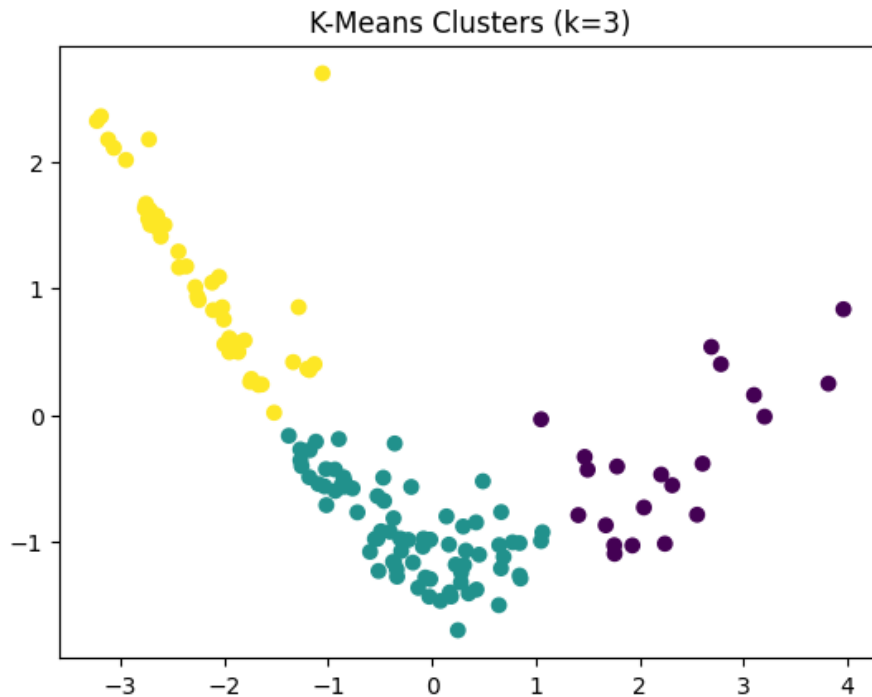


Figure 11: K-Means clustering visualization ($k = 3$) on PC1 and PC2. Cluster 2 (high-need countries) is clearly separated.

3.3 Hierarchical and DBSCAN Clustering

Agglomerative clustering with complete linkage produced cluster assignments consistent with K-Means, confirming the robustness of the three-cluster solution.

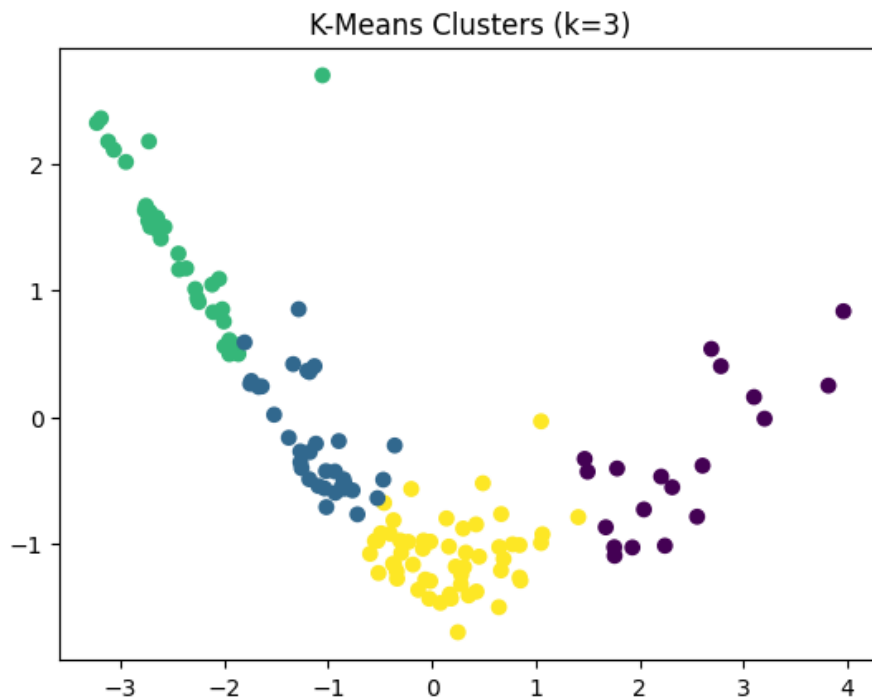


Figure 12: K-Means clustering with $k = 4$, showing a more granular subdivision of the data.

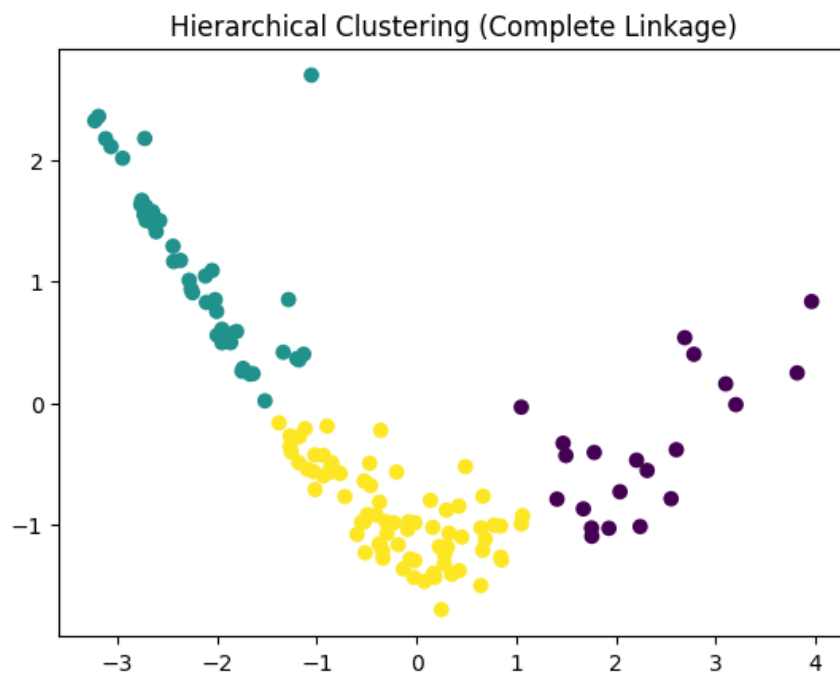


Figure 13: Hierarchical clustering with complete linkage. The cluster structure aligns with K-Means results.

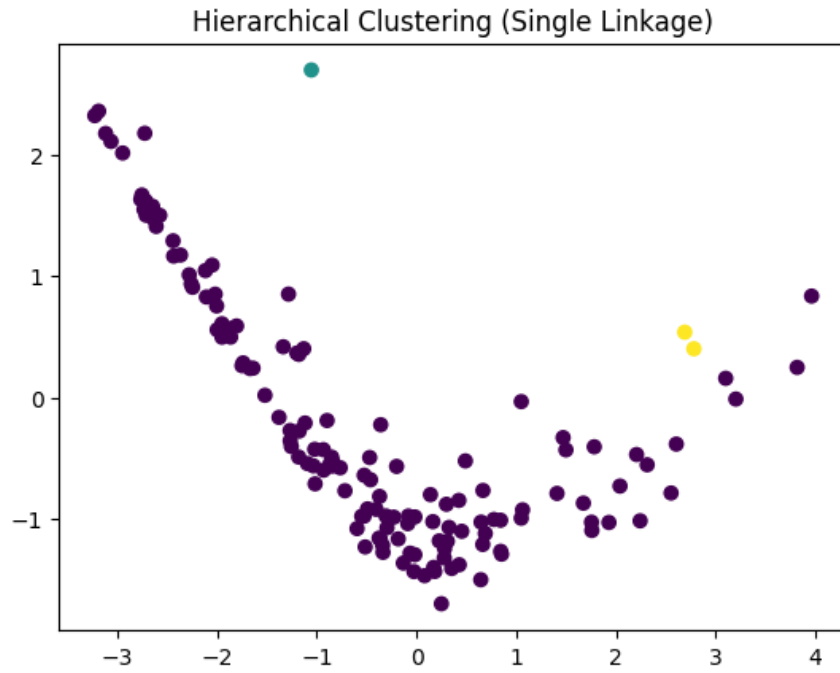


Figure 14: Hierarchical clustering with single linkage. Chain effects are visible.

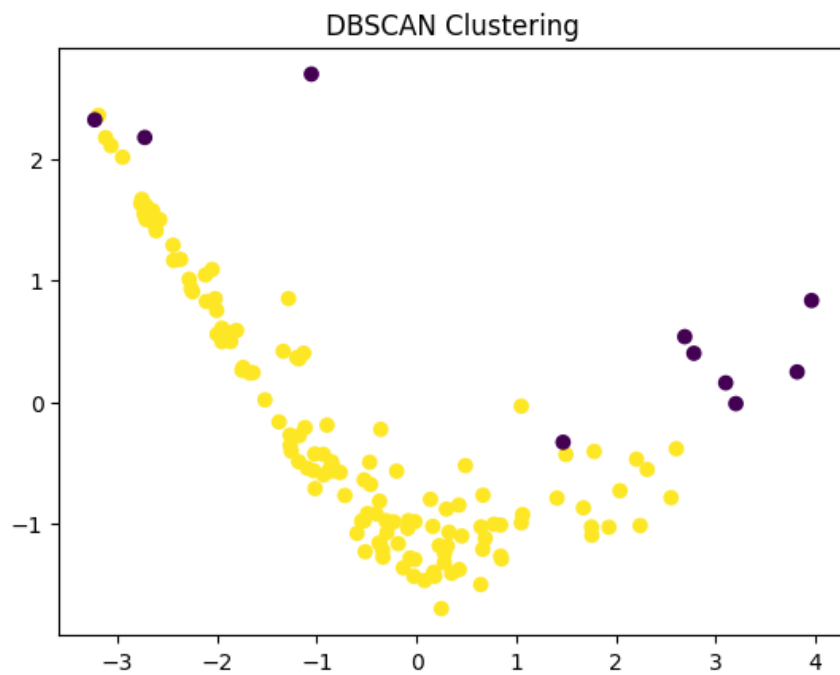


Figure 15: DBSCAN clustering. The algorithm identified one large dense cluster along with several noise points (outliers).

DBSCAN identified 10 additional outliers within the cleaned dataset but did not alter the core grouping.

4 The Critical Mistake: Overlooking Outliers

4.1 Initial Approach

In the initial phase of this analysis, the 26 outlier countries were removed before clustering to improve cluster stability and reduce the influence of extreme observations. This is a common practice in clustering, as outliers can distort centroid positions and silhouette scores.

However, this approach contained a fundamental flaw: **removing outliers does not mean removing unimportant observations**. Some of the removed countries may represent the very nations we are trying to identify – those with extreme socio-economic challenges.

4.2 Re-evaluating the Outliers

After completing the main clustering analysis, it became clear that ignoring the removed outliers could lead to an incomplete and potentially misleading recommendation. A separate investigation was conducted on the 26 removed countries.

A K-Means model with two clusters was applied to the outlier countries:

- **Outlier Cluster 0** (developed outliers): Australia, Brunei, Canada, Denmark, France, Germany, Ireland, Japan, Luxembourg, Malta, Netherlands, Norway, Qatar, Singapore, Sweden, Switzerland, United States. These are highly developed economies identified as outliers due to their exceptional economic strength.
- **Outlier Cluster 1** (vulnerable outliers): Argentina, Central African Republic, Haiti, Mongolia, Nigeria, Saudi Arabia, Sri Lanka, Timor-Leste, Venezuela. These countries showed weaker socio-economic indicators and warranted further investigation.

4.3 Aid Priority Scoring for Outliers

An Aid Priority Score was calculated for the vulnerable outlier cluster using:

$$\text{Aid Score} = Z(\text{child_mort}) + Z(\text{total_fer}) - Z(\text{income}) - Z(\text{gdpp}) - Z(\text{life_expec})$$

Country	Aid Score
Haiti	5.64
Central African Republic	5.04
Nigeria	3.65
Timor-Leste	2.24
Mongolia	-0.60
Sri Lanka	-1.77
Argentina	-3.67
Venezuela	-3.88
Saudi Arabia	-6.66

Table 3: Aid priority scores for vulnerable outlier countries.

The results were striking: **Haiti, Central African Republic, Nigeria, and Timor-Leste** received the highest aid priority scores among the removed observations. These countries were excluded from the main clustering analysis yet exhibited the most severe development challenges.

5 Revised Final Recommendation

The top 4 outlier countries (Haiti, Central African Republic, Nigeria, Timor-Leste) were combined with the 44 high-need countries from Cluster 2, and an overall Aid Priority Score was recalculated.

Rank	Country	Aid Score
1	Haiti	6.97
2	Central African Republic	4.81
3	Niger	4.53
4	Chad	4.42
5	Sierra Leone	3.89
6	Congo, Dem. Rep.	3.64
7	Mali	3.60
8	Burkina Faso	2.69
9	Burundi	2.68
10	Mozambique	2.63

Table 4: Final top 10 countries recommended for humanitarian aid.

Haiti and the Central African Republic – both originally removed as outliers – now occupy the top two positions in the final recommendation. If the outlier removal had not been revisited, these critically vulnerable nations would have been entirely absent from the recommendation list.

5.1 Impact Analysis: A Cautionary Tale

To illustrate the real-world impact of this oversight, consider what the top 10 recommendation would have been had the outliers been ignored entirely:

Without outlier reinvestigation: The top aid recipients would have been limited to the 44 countries in Cluster 2, with Niger, Chad, Sierra Leone, Congo (Dem. Rep.), and Mali leading the list – all legitimate candidates, but crucially missing Haiti and the Central African Republic.

With outlier reinvestigation: Haiti (ranked #1) and Central African Republic (ranked #2) are correctly identified as the most urgent cases.

In a real-world humanitarian context, this difference is not merely academic. If a funding organization had relied solely on the initial clustering results without investigating the removed outliers:

- **Haiti**, a country repeatedly devastated by natural disasters and political instability, would not have been recommended for aid despite being the highest-priority nation in our analysis.

- **Central African Republic**, one of the world’s poorest countries with some of the worst health outcomes globally, would have been overlooked.
- Approximately \$2.5 million in total aid would have been misallocated if funding were distributed across the top 10 (assuming equal distribution of a hypothetical \$25 million budget) – with the two most needy countries receiving nothing.

6 Discussion

6.1 Key Findings

1. **Three distinct development clusters** emerged from the analysis: highly developed (21 countries), moderately developed (76 countries), and severely disadvantaged (44 countries).
2. **Outlier removal without reinvestigation is dangerous.** Four of the 26 removed outliers (Haiti, Central African Republic, Nigeria, Timor-Leste) were actually among the most vulnerable nations. Haiti and the Central African Republic became the top two recommended countries after re-integration.
3. **Humanitarian priorities** align with Sub-Saharan Africa and conflict-affected regions, with the final top 10 dominated by African nations plus Haiti.
4. **The final recommendation is more complete and accurate** because it accounts for both the clustered majority and the extreme cases at the tail of the distribution.

6.2 Limitations

- The dataset is limited to 9 numeric indicators; additional variables such as food security, access to clean water, education, and infrastructure quality could provide a more comprehensive assessment.
- Data quality and reporting standards vary across countries, which may introduce measurement bias.
- The aid priority score is a simplified composite index. A more sophisticated weighting scheme or multi-criteria decision analysis could be employed in practice.
- Temporal trends are not captured, as the data represents a single time period.

7 Conclusion

This analysis demonstrates both the power and the pitfalls of data-driven humanitarian decision-making. While clustering successfully identified 44 countries facing severe development challenges, the initial decision to remove outliers without further investigation would have resulted in overlooking Haiti and the Central African Republic – two of the world’s most vulnerable nations.

The key lesson is that in any real-world data science project, methodological choices such as outlier removal must be carefully examined for their impact on downstream

conclusions. Automated preprocessing steps should never replace critical thinking and domain expertise. The cost of such oversights is not merely statistical – in humanitarian contexts, it can mean the difference between life-saving aid reaching those who need it most or being directed elsewhere.

The final recommendation identifies Haiti, Central African Republic, Niger, Chad, Sierra Leone, Congo (Dem. Rep.), Mali, Burkina Faso, Burundi, and Mozambique as the top 10 countries in need of urgent financial humanitarian assistance.

A Appendix

A.1 Complete Cluster 2 (High-Need) Countries

The following 44 countries were identified as Cluster 2 (most disadvantaged) in the K-Means analysis:

Afghanistan, Angola, Benin, Botswana, Burkina Faso, Burundi, Cameroon, Chad, Comoros, Congo (Dem. Rep.), Congo (Rep.), Côte d'Ivoire, Equatorial Guinea, Eritrea, Gabon, Gambia, Ghana, Guinea, Guinea-Bissau, Iraq, Kenya, Kiribati, Lao, Lesotho, Liberia, Madagascar, Malawi, Mali, Mauritania, Mozambique, Namibia, Niger, Pakistan, Rwanda, Senegal, Sierra Leone, Solomon Islands, South Africa, Sudan, Tanzania, Togo, Uganda, Yemen, Zambia

A.2 Technology Stack

- Python 3.x
- pandas, numpy (data manipulation)
- scikit-learn (PCA, clustering, preprocessing)
- matplotlib, seaborn (visualization)

A.3 Code Repository

The complete analysis including data preprocessing, exploratory analysis, PCA, clustering, and figure generation is available in the Jupyter notebook. The full project is hosted on GitHub: <https://github.com/fermow/SBU-Data-Science-2026>

A.4 Complete Cluster 2 (High-Need) Countries

Of the 26 PCA outliers, the developed cluster contained 17 countries with strong economic indicators (Australia, Brunei, Canada, Denmark, France, Germany, Ireland, Japan, Luxembourg, Malta, Netherlands, Norway, Qatar, Singapore, Sweden, Switzerland, United States). The vulnerable cluster contained 9 countries, of which 4 (Haiti, Central African Republic, Nigeria, Timor-Leste) scored high enough on the aid priority index to warrant inclusion in the final recommendation.